

Generation Interconnection System Impact Study Report

For Queue Project Q20-1476

Country Club 2 Tap 138 kV

September 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1476. The Interconnection Customer proposes a 109.6 MW battery energy storage facility interconnecting at the Country Club 2 Tap 138 kV substation in the New Jersey area. The request was received on May 13, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 508 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$652,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Montour Station 500/345 kV Transformer Bank
Contingency: Coronet - Westwood Generation Llc 500 KV
Loading: 101.0% (Rating: 1,439 MVA)
DFAX: 6.3%
Pre-project loading: 100.6%
2. Facility: Sunbury 500-345 kV Autotransformer
Contingency: Westwood Generation Llc - Coronet 500 KV
Loading: 100.8% (Rating: 1,416 MVA)
DFAX: 6.2%
Pre-project loading: 100.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montour Station 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$50,995,000	1% (\$444,000)
Sunbury 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$21,881,000	1% (\$208,000)
	Total allocated to project		\$652,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.